

Exploring the role of retinal cryptochromes in avian navigation

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Birds rely on their eyes to collect not only visual, but also magnetic cues from the Earth's magnetic field to orient and navigate over long distances. A class of blue-light sensitive flavoproteins, the cryptochromes, are suggested as possible magnetoreceptive molecule candidates. Due to its biochemical properties and localization, Cryptochrome 4 (Cry4) is discussed as the most likely primary magnetoreceptor. To study its distribution in tissue on single-cell and subcellular level, immunohistochemistry can be used. However, for wild bird species, no commercial antibodies are available. We have previously generated European robin (*Erithacus rubecula*)-specific antibodies that showed Cry4 localization to the outer segments of double cones and long-wavelength single cones, but it is not known if this localization is preserved in other bird species.

Our aim was to generate a polyclonal antibody that reliably detects Cry4 in different bird species. We immunized a rabbit with the full-length sequence of blackcap (*Sylvia atricapilla*) and quail (*Coturnix coturnix*) Cry4. After affinity purification of the resulting serum, we validated the antibodies in QNR/K2 cells transfected with the European robin Cry4 sequence. In immunocytochemical stainings, each antibody candidate was evaluated in combination with the monoclonal Cry4 marker 3C2.

All QNR/K2 cells showed a signal from both antibodies simultaneously, except for non-transfected cells showing no signal from either antibody. This finding suggests strong specificity of the developed antibodies. In Western blots of lysate from transfected QNR/K2 cells, we detected single characteristic bands at the expected molecular mass of Cry4 for each antibody.

Our future steps include using the validated antibodies to characterize and compare Cry4 distribution *in situ* in the retina of multiple night-migratory bird species. While molecular interactions of Cry4 have been studied in transfected cell lines, they might differ in the actual retinal cells. Since it is difficult to culture isolated photoreceptors, we intend to use retinal explants or retinal organoids as model system for investigation.

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